Mounted at /content/gdrive2
Drive root: /content/gdrive2
GlucoIris folder: /content/gdrive2/MyDrive/GlucoIris
Checkpoint exists: True
Output directory: /content/gdrive2/MyDrive/GlucoIris/final_vit_refinement
==
GLUCOIRIS — FIVE-CLASS ViT REFINEMENT
==
PyTorch: 2.11.0+cu130
Device: cuda
GPU: Tesla T4

APTOS dataset located.
CSV: /content/aptos/train.csv
Image files indexed: 5590

Total labelled images: 3662

Full dataset class distribution:
diagnosis
0 1805
1 370
2 999
3 193
4 295
Name: count, dtype: int64

Checkpoint model name: vit_small_patch16_224.augreg_in21k_ft_in1k
Image size: 224
Original validation QWK: 0.8844844192214797

Existing ViT checkpoint loaded strictly.
Architecture: ViT-Small/16-like
Parameters: 21,667,589

Stratified split sizes:
Train: 2563
Validation: 549
Test: 550

Train distribution:
Grade 0 (No DR): 1263 / 2563 (49.28%)
Grade 1 (Mild): 259 / 2563 (10.11%)
Grade 2 (Moderate): 699 / 2563 (27.27%)

Grade 3 (Severe): 135 / 2563 (5.27%)
Grade 4 (Proliferative DR): 207 / 2563 (8.08%)

Validation distribution:
Grade 0 (No DR): 271 / 549 (49.36%)
Grade 1 (Mild): 55 / 549 (10.02%)
Grade 2 (Moderate): 150 / 549 (27.32%)
Grade 3 (Severe): 29 / 549 (5.28%)
Grade 4 (Proliferative DR): 44 / 549 (8.01%)

Test distribution:
Grade 0 (No DR): 271 / 550 (49.27%)
Grade 1 (Mild): 56 / 550 (10.18%)
Grade 2 (Moderate): 150 / 550 (27.27%)
Grade 3 (Severe): 29 / 550 (5.27%)
Grade 4 (Proliferative DR): 44 / 550 (8.00%)

Weighted sampling enabled.
Augmentation enabled for training only.

Baseline model on the recreated validation split:
Accuracy: 0.7887
Macro F1: 0.6156
QWK: 0.8823
embeddings: 371,328 params, LR=1.0e-06
early_blocks: 14,195,712 params, LR=2.0e-06
late_blocks: 7,097,856 params, LR=1.0e-05
other: 768 params, LR=3.0e-06
head: 1,925 params, LR=5.0e-05

Epoch 1/12
Train loss=0.4810 | Train accuracy=0.8646
Val loss=0.6036 | Val accuracy=0.7942 | Macro F1=0.6413 | QWK=0.8779
Saved best validation-accuracy checkpoint.
Saved best validation-QWK checkpoint.

Epoch 2/12
Train loss=0.4159 | Train accuracy=0.8787
Val loss=0.6159 | Val accuracy=0.8033 | Macro F1=0.6582 | QWK=0.8811
Saved best validation-accuracy checkpoint.
Saved best validation-QWK checkpoint.

Epoch 3/12
Train loss=0.3726 | Train accuracy=0.8911

Val loss=0.6245 | Val accuracy=0.8033 | Macro F1=0.6473 | QWK=0.8842
Saved best validation-QWK checkpoint.

Epoch 4/12
Train loss=0.3288 | Train accuracy=0.9153
Val loss=0.6244 | Val accuracy=0.8124 | Macro F1=0.6569 | QWK=0.8900
Saved best validation-accuracy checkpoint.
Saved best validation-QWK checkpoint.

Epoch 5/12
Train loss=0.3055 | Train accuracy=0.9231
Val loss=0.6785 | Val accuracy=0.7851 | Macro F1=0.6313 | QWK=0.8862

Epoch 6/12
Train loss=0.3022 | Train accuracy=0.9263
Val loss=0.7034 | Val accuracy=0.7923 | Macro F1=0.6388 | QWK=0.8847

Epoch 7/12
Train loss=0.2697 | Train accuracy=0.9384
Val loss=0.6853 | Val accuracy=0.7960 | Macro F1=0.6343 | QWK=0.8868

Epoch 8/12
Train loss=0.2529 | Train accuracy=0.9423
Val loss=0.6863 | Val accuracy=0.8033 | Macro F1=0.6367 | QWK=0.8757

Early stopping: validation accuracy plateaued.

Training complete.
Best-accuracy checkpoint:
/content/gdrive2/MyDrive/GlucoIris/final_vit_refinement/glucoiris_vit_refined_best_acc.pt
Best-QWK checkpoint:
/content/gdrive2/MyDrive/GlucoIris/final_vit_refinement/glucoiris_vit_refined_best_qwk.pt

==========
SELECTED MODEL — VALIDATION METRICS
==========
Accuracy: 0.8124
Macro F1: 0.6569
QWK: 0.8900

==========
FINAL TEST RESULTS
==========
Test accuracy: 0.8073

Test macro F1: 0.6498
Test QWK: 0.8786

Confusion matrix:

	No DR	Mild	Moderate	Severe	Proliferative DR
No DR	264	6	1	0	0
Mild	4	32	18	0	2
Moderate	3	12	114	15	6
Severe	0	0	7	14	8
Proliferative DR	1	1	15	7	20

Per-class report:

	precision	recall	f1-score	support
No DR	0.9706	0.9742	0.9724	271
Mild	0.6275	0.5714	0.5981	56
Moderate	0.7355	0.7600	0.7475	150
Severe	0.3889	0.4828	0.4308	29
Proliferative DR	0.5556	0.4545	0.5000	44
accuracy			0.8073	550
macro avg	0.6556	0.6486	0.6498	550
weighted avg	0.8077	0.8073	0.8066	550

Referable DR (grades 2–4):
TP: 206
FN: 17
TN: 306
FP: 21
Sensitivity: 0.9238
Specificity: 0.9358

Saved training history:
/content/gdrive2/MyDrive/GlucoIris/final_vit_refinement/training_history.csv
Saved results: /content/gdrive2/MyDrive/GlucoIris/final_vit_refinement/final_test_results.json
Saved confusion matrix:
/content/gdrive2/MyDrive/GlucoIris/final_vit_refinement/test_confusion_matrix.csv
Original checkpoint left unchanged: /content/gdrive2/MyDrive/GlucoIris/glucoiris_vit_best.pt

ALL DONE.